# PaySim Fraud Detection

## Final Results

Models: XGBoost · Isolation Forest · Autoencoder · Ensemble

This report summarises detection performance on the held-out test split.

Metrics shown match the dashboard exactly.

## Dataset Summary

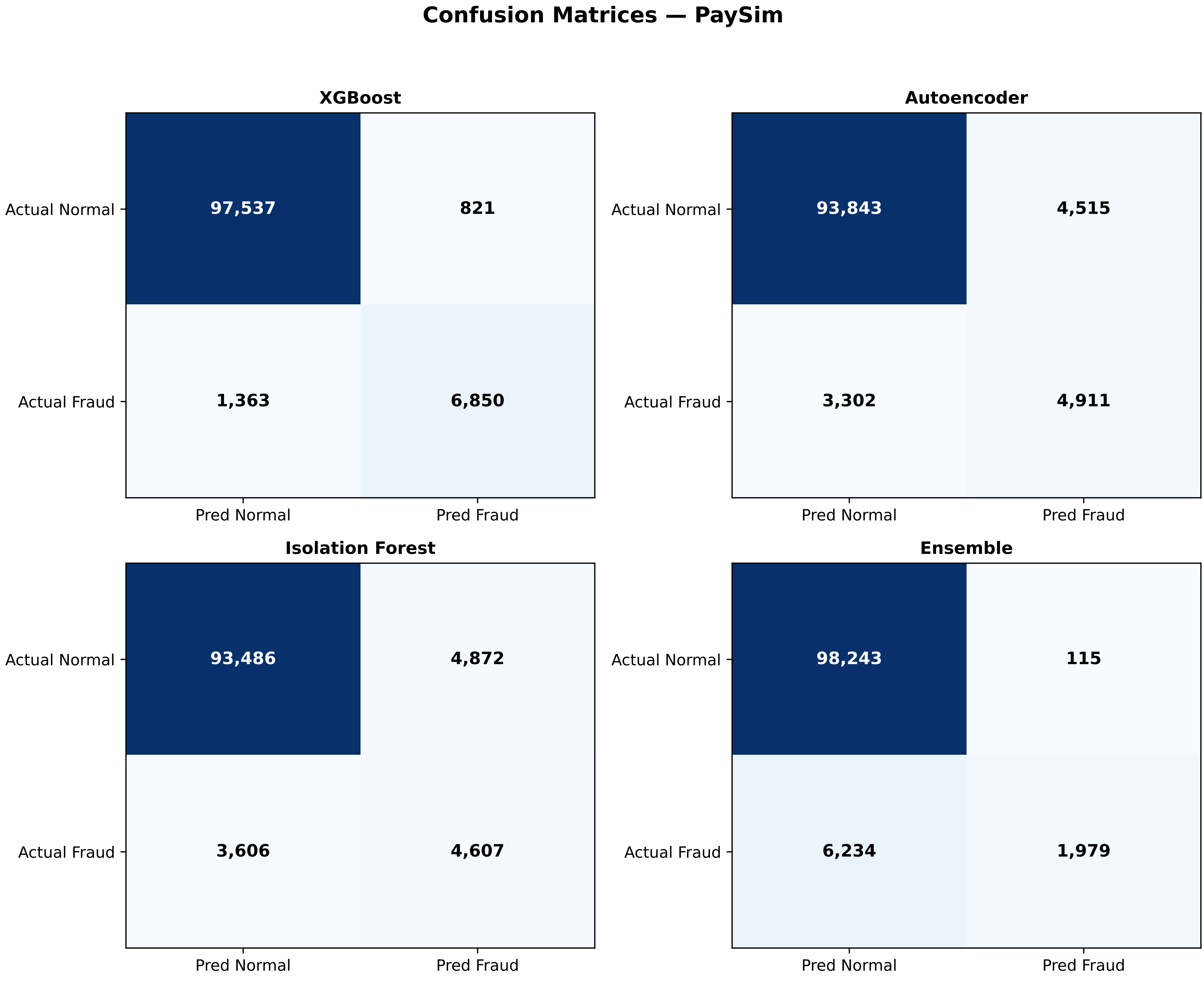
| Field | Value |
| --- | --- |
| Dataset | PaySim |
| Total Samples | 500,000 |
| Normal Samples | 491,787 |
| Fraud Samples | 8,213 |
| Fraud Ratio | 1.6426% |
| Number of Features | 11 |
| Test Samples | 106,571 |
| Test Fraud | 8,213 |
| Contamination | 0.016426 |

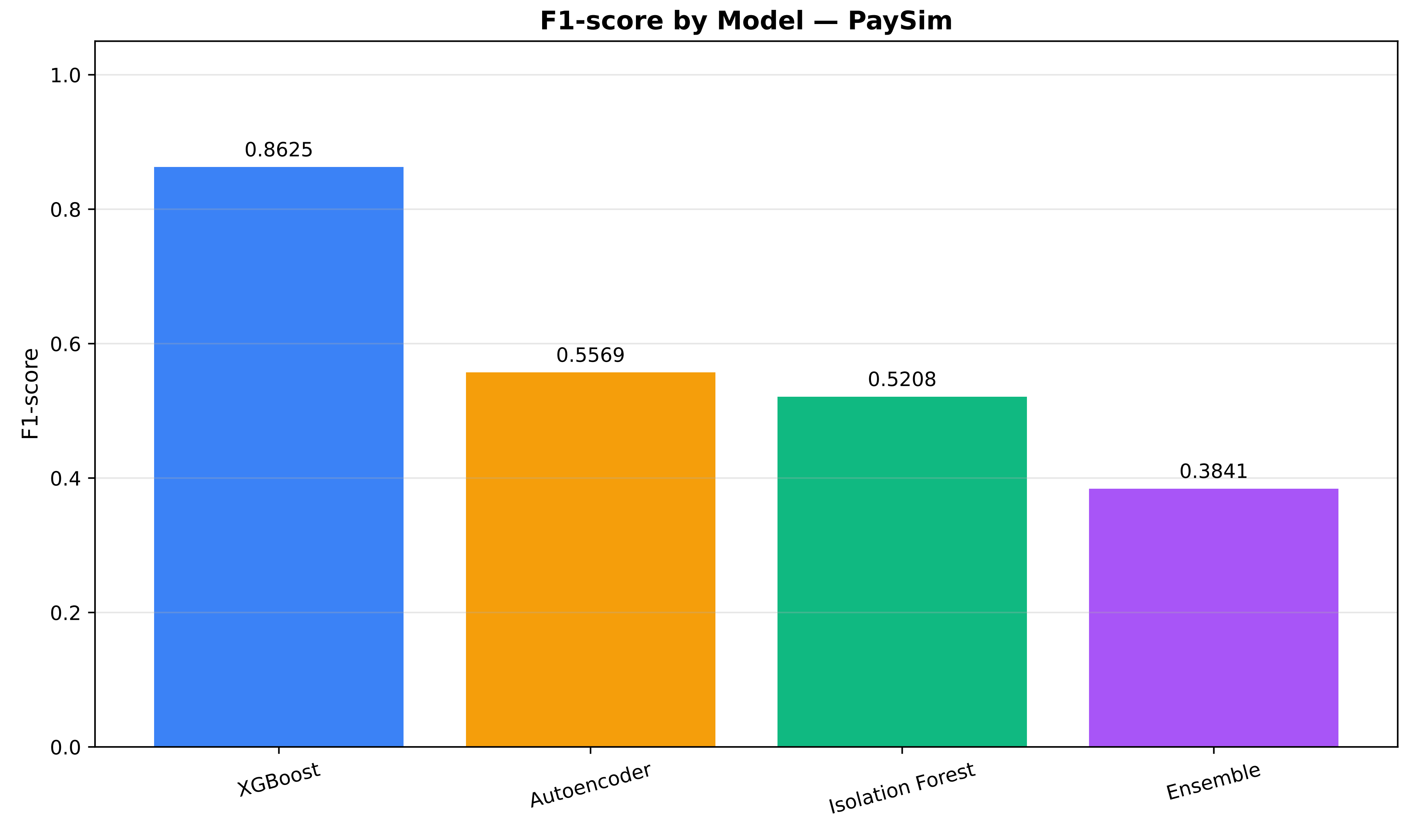
## Hyperparameters

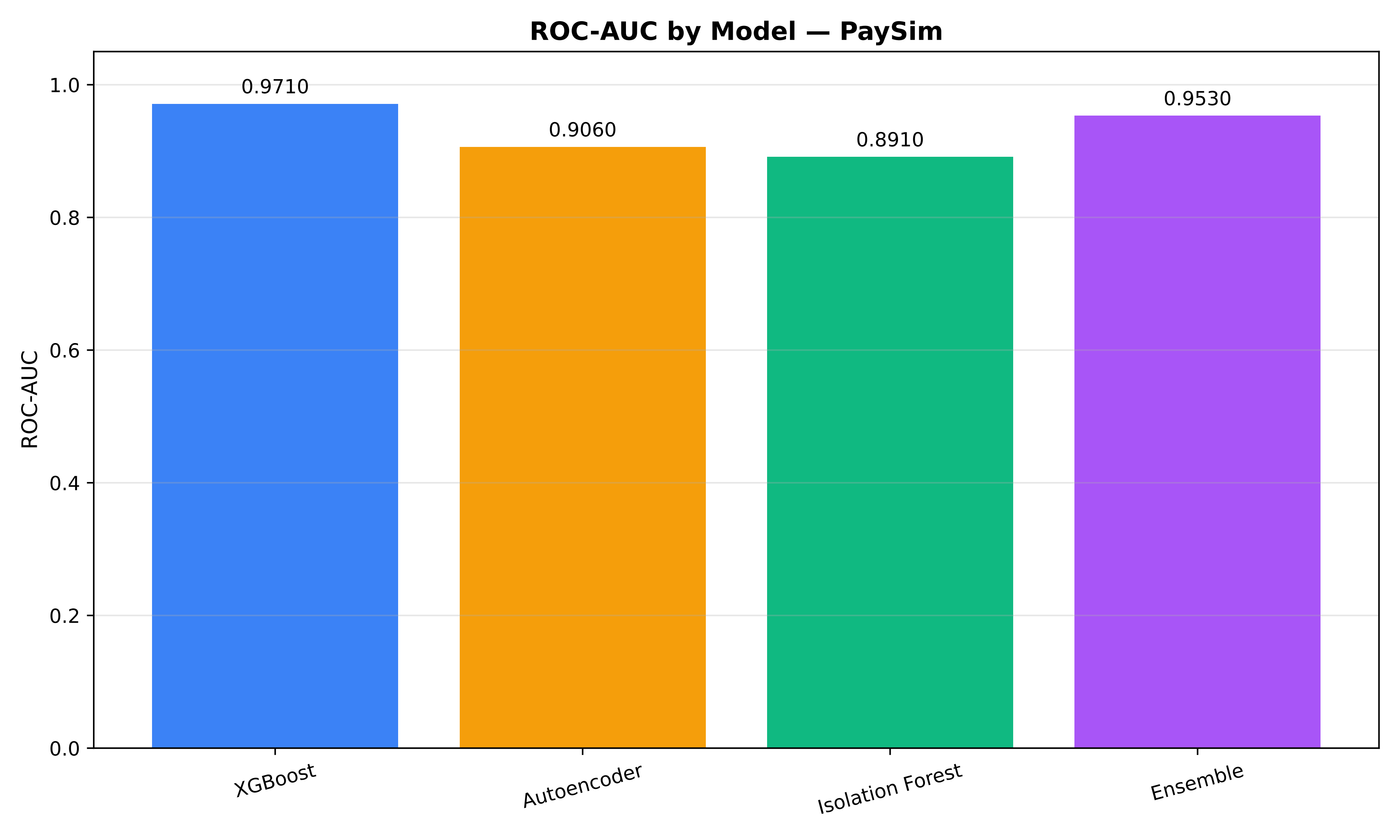
| Model | Hyperparameters |
|---|---|
| XGBoost | n_estimators=300, max_depth=6, learning_rate=0.1, scale_pos_weight=60<br>eval_metric='logloss', random_state=42 |
| Isolation Forest | n_estimators=300, max_samples=0.8, contamination=0.016426<br>random_state=42 |
| Autoencoder | encoding_dim=8, epochs=10, batch_size=1024<br>threshold=percentile(1-contamination) |
| Ensemble | Mean of min-max normalized scores: XGBoost + Isolation Forest +<br>Autoencoder. Threshold at training-time percentile. |

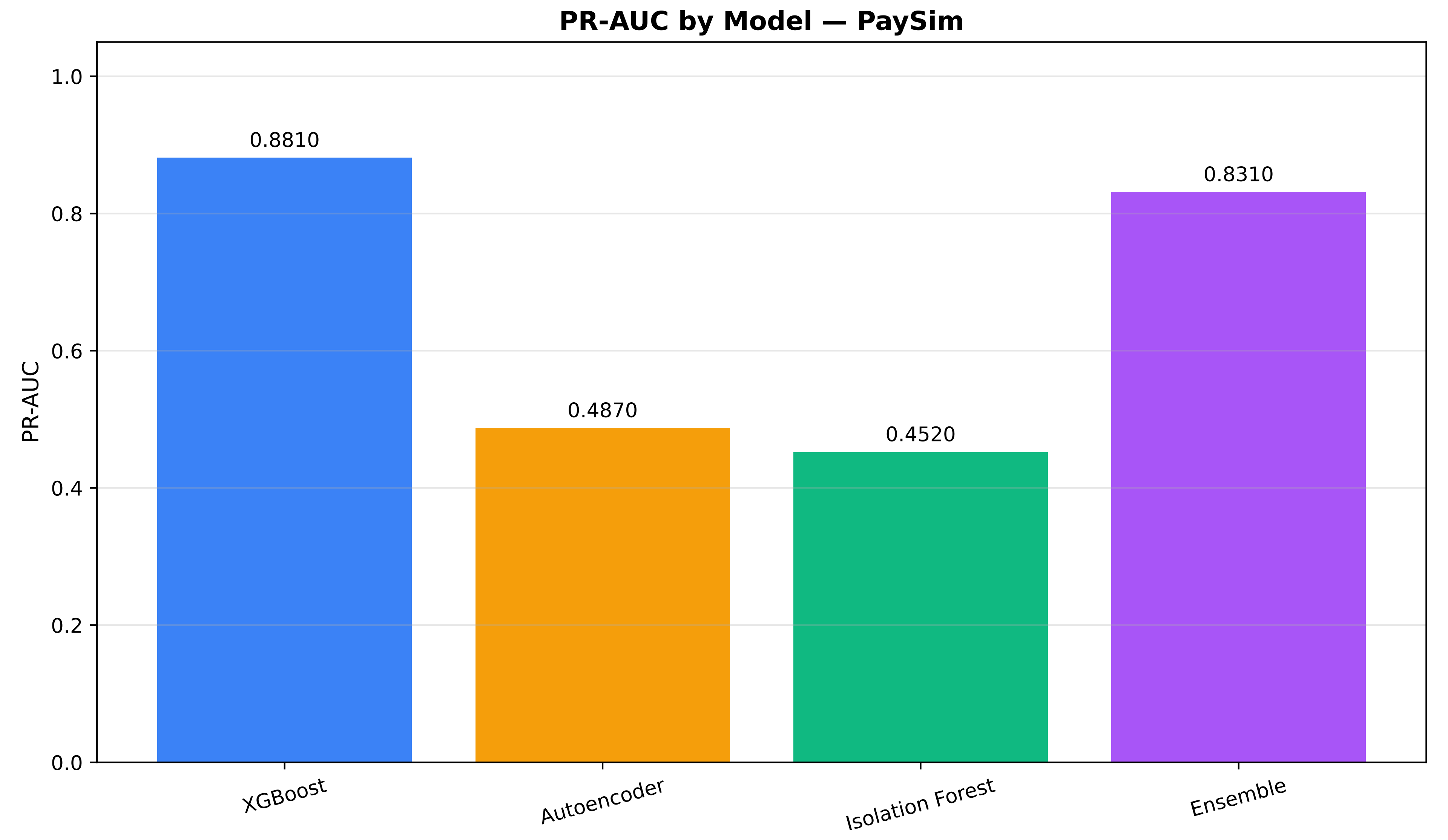
## Per-Model Evaluation

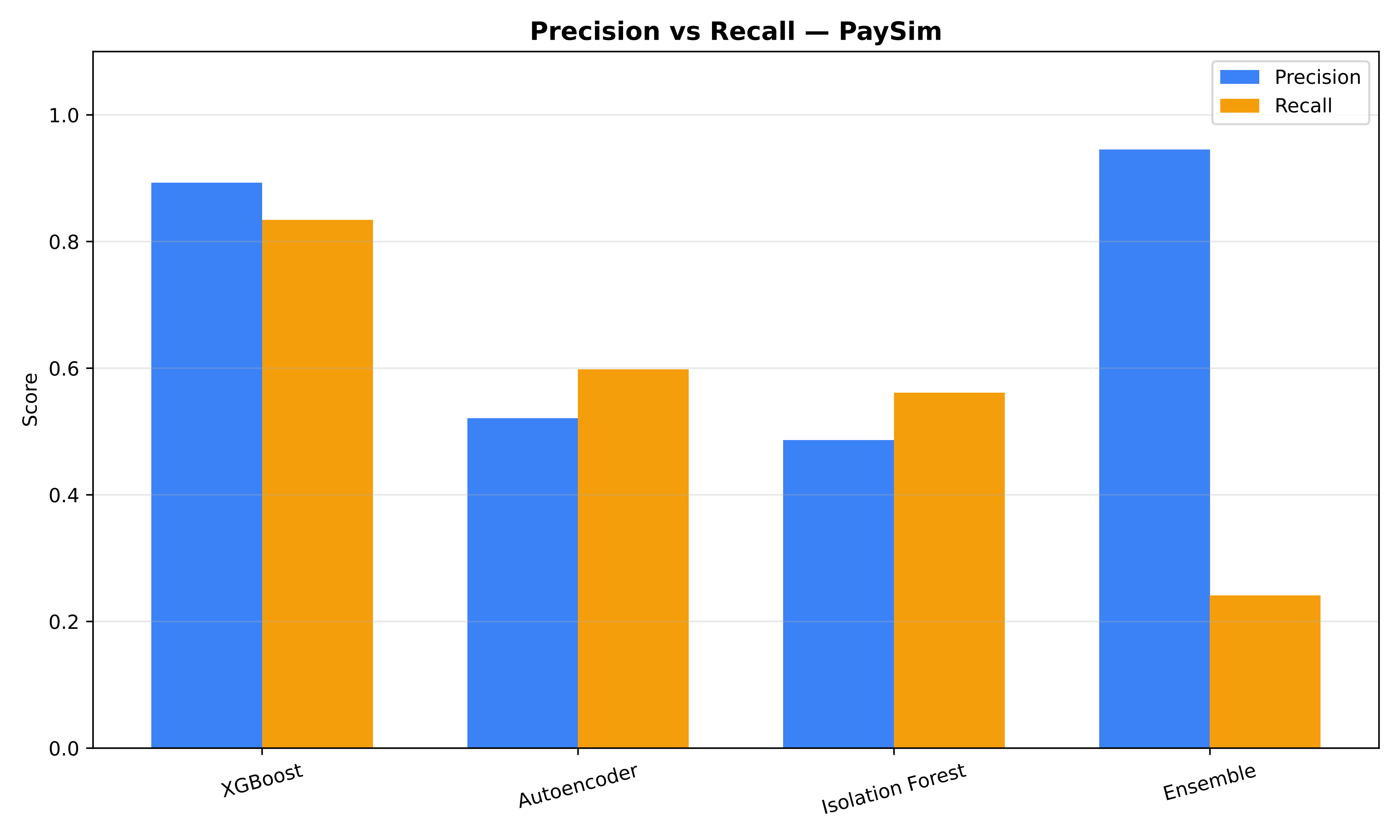
| Rank | Model | Precision | Recall | F1-score | ROC-AUC | PR-AUC | TN | FP | FN | TP |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | XGBoost | 0.8930 | 0.8340 | 0.8625 | 0.9710 | 0.8810 | 97537 | 821 | 1363 | 6850 |
| 2 | Autoencoder | 0.5210 | 0.5980 | 0.5569 | 0.9060 | 0.4870 | 93843 | 4515 | 3302 | 4911 |
| 3 | Isolation Forest | 0.4860 | 0.5610 | 0.5208 | 0.8910 | 0.4520 | 93486 | 4872 | 3606 | 4607 |
| 4 | Ensemble | 0.9450 | 0.2410 | 0.3841 | 0.9530 | 0.8310 | 98243 | 115 | 6234 | 1979 |


Confusion Matrices — PaySim
XGBoost
Actual Normal
Actual Fraud
97,537
821
1,363
6,850
Pred Normal
Pred Fraud
Autoencoder
Actual Normal
Actual Fraud
93,843
4,515
3,302
4,911
Pred Normal
Pred Fraud
Isolation Forest
Actual Normal
Actual Fraud
93,486
4,872
3,606
4,607
Pred Normal
Pred Fraud
Ensemble
Actual Normal
Actual Fraud
98,243
115
6,234
1,979
Pred Normal
Pred Fraud


F1-score by Model — PaySim
F1-score
1.0
0.8
0.6
0.4
0.2
0.0
0.8625
0.5569
0.5208
0.3841
XGBoost
Autoencoder
Isolation Forest
Ensemble


ROC-AUC by Model — PaySim
ROC-AUC
0.0
0.2
0.4
0.6
0.8
1.0
0.9710
0.9060
0.8910
0.9530
XGBoost
Autoencoder
Isolation Forest
Ensemble


PR-AUC by Model — PaySim
PR-AUC
0.0
0.2
0.4
0.6
0.8
1.0
0.8810
0.4870
0.4520
0.8310
XGBoost
Autoencoder
Isolation Forest
Ensemble


Precision vs Recall — PaySim
Precision
Recall
Score
1.0
0.8
0.6
0.4
0.2
0.0
XGBoost
Autoencoder
Isolation Forest
Ensemble